

- This exam contains 12 pages (including the front and back of this cover page) and 14 problems. Check to see if any pages are missing.
- This is the final exam and will count for 20% of your final grade. You have 120 minutes (3:30 pm - 5:30 pm) to complete this exam. You may use one index card of notes and a calculator on this exam, but no other outside sources may be consulted.

The following rules apply:

- Every problem on this exam, other than true-or-false problems, requires work of some form. An incorrect answer supported by substantially correct calculations and explanations will receive partial credit. **A correct answer that has no supporting work will not receive full credit.**
- Organize your work, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little to no credit.
- There are 100 points total in this exam, as well as 1 bonus questions worth a total of 5 possible points.

Page:	3	4	5	6	7	8	9	10	11	Total
Points:	10	10	10	10	8	8	14	20	10	100
Score:										

Bonus:	
--------	--

1. Mark each statement as ☐ True or ☐ False by completely shading in the box corresponding to your answer.

For readability in the key, correct answers are colored in red, e.g. **Correct Answer**, while incorrect answers are marked with a strike, e.g. ~~Incorrect Answer~~.

2 pts

- (a) The conic section represented by the equation $4x^2 + 16x - 16y^2 + 64y + 72 = 0$ is a parabola.

☒ True

☐ False

2 pts

- (b) The conic section represented by the equation $3x^2 + 5xy + 4y^2 + 1 = 0$ is an ellipse.

☒ True

☐ False

2 pts

- (c) An explicit formula for the arithmetic sequence (starting at a_1) given by $\{-2/3, -1, -4/3, \dots\}$ is $a_n = -(1/3)n - 1/3$.

☒ True

☐ False

2 pts

- (d) In summation notation, the geometric sum $1 + 4 + 16 + 64 + 256 + 1024 + 4096 + 16384$ can be expressed as $\sum_{n=0}^8 4^n$.

☒ True

☐ False

2 pts

- (e) The sequence $\{-1.4, 7, -35, 175, -875, \dots\}$ is geometric.

☒ True

☐ False

10 pts

2. Write the equation of the ellipse in standard form, and identify the end points of the major and minor axes as well as the foci.

$$25x^2 - 50x + 16y^2 - 96y - 231 = 0$$

Equation in Standard Form :	$\frac{(x-1)^2}{4^2} + \frac{(y-3)^2}{5^2} = 1$
End Points of Major Axis :	$(1, -2), (1, 8)$
End Points of Minor Axis :	$(-3, 3), (5, 3)$
Foci :	$(1, 0), (1, 6)$

To begin, we re-write the equation given in the standard form

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1.$$

To do so, we will need to complete the square for both the terms involving x and the terms involving y . This may be accomplished as follows:

$$\begin{aligned}
 25x^2 - 50x + 16y^2 - 96y - 231 &= 0 \\
 25(x^2 - 2x) + 16(y^2 - 6y) - 231 &= 0 \\
 25(x^2 - 2x + 1 - 1) + 16(y^2 - 6y + 9 - 9) - 231 &= 0 \\
 25(x^2 - 2x + 1) - 25 + 16(y^2 - 6y + 9) - 144 - 231 &= 0 \\
 25(x-1)^2 + 16(y-3)^2 - 400 &= 0 \\
 25(x-1)^2 + 16(y-3)^2 &= 400 \\
 \frac{25(x-1)^2}{400} + \frac{16(y-3)^2}{400} &= 1 \\
 \frac{(x-1)^2}{16} + \frac{(y-3)^2}{25} &= 1 \\
 \frac{(x-1)^2}{4^2} + \frac{(y-3)^2}{5^2} &= 1
 \end{aligned}$$

Now, with the standard form of the equation in hand, we can determine the rest of the information requested by the problem. Since $5 > 4$, it follows that the major axis is parallel to the y -axis. So, the coordinates of the vertices (i.e. the end points of the major axis) are $(h, k \pm b)$, i.e. $(1, 3 - 5) = (1, -2)$ and $(1, 3 + 5) = (1, 8)$. Similarly, the coordinates of the co-vertices (i.e. the end points of the minor axis) are $(h \pm a, k)$, i.e. $(1 - 4, 3) = (-3, 3)$ and $(1 + 4, 3) = (5, 3)$. Finally, computing $c = \sqrt{a^2 + b^2} = \sqrt{5^2 - 4^2} = 3$, the coordinates of the foci are $(h, k \pm c)$, i.e. $(1, 3 - 3) = (1, 0)$ and $(1, 3 + 3) = (1, 6)$.

10 pts

3. Write the equation of the hyperbola in standard form, and identify the vertices and the foci.

$$16x^2 - 32x - 9y^2 - 36y - 164 = 0$$

Equation in Standard Form : $\frac{(x-1)^2}{3^2} - \frac{(y+2)^2}{4^2} = 1$

Vertices : $(-2, -2), (4, -2)$

Foci : $(-4, -2), (6, -2)$

To begin, we re-write the equation given in the standard form

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1.$$

To do so, we will need to complete the square for both the terms involving x and the terms involving y . This may be accomplished as follows:

$$\begin{aligned} 16x^2 - 32x - 9y^2 - 36y - 164 &= 0 \\ 16(x^2 - 2x) - 9(y^2 + 4y) - 164 &= 0 \\ 16(x^2 - 2x + 1 - 1) - 9(y^2 + 4y + 4 - 4) - 164 &= 0 \\ 16(x^2 - 2x + 1) - 16 - 9(y^2 + 4y + 4) + 36 - 164 &= 0 \\ 16(x-1)^2 - 9(y+2)^2 - 144 &= 0 \\ 16(x-1)^2 - 9(y+2)^2 &= 144 \\ \frac{16(x-1)^2}{144} - \frac{9(y+2)^2}{144} &= 1 \\ \frac{(x-1)^2}{9} - \frac{(y+2)^2}{16} &= 1 \\ \frac{(x-1)^2}{3^2} - \frac{(y+2)^2}{4^2} &= 1 \end{aligned}$$

Now, with the standard form of the equation in hand, we can determine the rest of the information requested by the problem. Since the term involving x 's is positive and the term involving the y 's is negative, it follows that the transverse axis is parallel to the x -axis. So, the coordinates of the vertices are $(h \pm a, k)$, i.e. $(1 - 3, -2) = (-2, -2)$ and $(1 + 3, -2) = (4, -2)$. Finally, computing $c = \sqrt{a^2 + b^2} = \sqrt{3^2 + 4^2} = 5$, the coordinates of the foci are $(h \pm c, k)$, i.e. $(1 - 5, -2) = (-4, -2)$ and $(1 + 5, -2) = (6, -2)$.

10 pts

4. Rewrite the given equation in standard form, and then determine the vertex, focus, directrix, and axis of symmetry of the parabola.

$$2y^2 + 8y + 8x - 24 = 0$$

$$\text{Equation in Standard Form : } \underline{(y + 2)^2 = 4(-1)(x - 4)}$$

$$\text{Vertex : } \underline{(4, -2)}$$

$$\text{Directrix : } \underline{x = 5}$$

$$\text{Axis of Symmetry : } \underline{y = -2}$$

To begin, we re-write the equation given in the standard form

$$(y - k)^2 = 4p(x - h).$$

To do so, we will need to complete the square for the terms involving y . This may be accomplished as follows:

$$2y^2 + 8y + 8x - 24 = 0$$

$$2(y^2 + 4y) + 8x - 24 = 0$$

$$2(y^2 + 4y + 4 - 4) + 8x - 24 = 0$$

$$2(y^2 + 4y + 4) - 8 + 8x - 24 = 0$$

$$2(y + 2)^2 + 8x - 32 = 0$$

$$2(y + 2)^2 = -8x + 32$$

$$(y + 2)^2 = -4x + 16$$

$$(y + 2)^2 = -4(x - 4)$$

Now, with the standard form of the equation in hand, we can determine the rest of the information requested by the problem. As always, the coordinate of the vertex is $(h, k) = (4, -2)$. Since the term involving y 's is squared and $p = -1 < 0$, it follows that the parabola opens to the left. So, the coordinate of the focus is $(h + p, k) = (4 + (-1), -2) = (3, -2)$ and the directrix is $x = h - p$, i.e. $x = 4 - (-1) = 5$. Finally, the axis of the symmetry is $y = k$, i.e. $y = -2$.

3 pts

5. Write the first six terms of the sequence.

$$a_1 = -2, \quad a_2 = 5, \quad a_n = a_{n-2}(4 - a_{n-1})$$

Using the recursive definition of the sequence we are given, we compute

$$a_3 = a_1(4 - a_2) = (-2)(4 - (5)) = 2$$

$$a_4 = a_2(4 - a_3) = (5)(4 - (2)) = 10$$

$$a_5 = a_3(4 - a_4) = (2)(4 - (10)) = -12$$

$$a_6 = a_4(4 - a_5) = (10)(4 - (-12)) = 160$$

So, the first six terms of the sequence are

$$-2, 5, 2, 10, -12, 160$$

5 pts

6. Write the first five terms of the arithmetic sequence given the following two terms

$$a_{11} = -185, \quad a_{31} = -485$$

Recall that the recursive formula for an arithmetic sequence is $a_n = a_{n-k} + kd$. Using what we have, this gives

$$a_{31} = a_{11} + (31 - 11)d$$

$$-485 = -185 + 20d$$

$$-300 = 20d$$

$$d = -15$$

So, $a_1 = a_{11} - 10d = -185 - 10(-15) = -35$. Using the explicit formula for an arithmetic sequence, $a_n = a_1 + (n - 1)d$, this gives

$$a_2 = -50 \quad a_3 = -65$$

$$a_4 = -80 \quad a_5 = -95$$

So, the first five terms of the arithmetic sequence are

$$-35, -50, -65, -80, -95$$

3 pts

7. Starting with a_1 , write the first five terms of the geometric sequence

$$a_n = -2(3)^{n-1}$$

Using the explicit definition of the geometric sequence we are given, we compute

$$a_1 = -2(3)^{1-1} = -2$$

$$a_2 = -2(3)^{2-1} = -6$$

$$a_3 = -2(3)^{3-1} = -18$$

$$a_4 = -2(3)^{4-1} = -54$$

$$a_5 = -2(3)^{5-1} = -162$$

So, the first six terms of the sequence are $-2, -6, -18, -54, -162$

5 pts

8. Find the 11^{th} term in the arithmetic sequence below.

$$\{-10, -13, -16, \dots\}$$

Since this is an arithmetic sequence, we begin by computing the common difference:

$$d = (-13) - (-10) = -3.$$

So, since the explicit formula for an arithmetic sequence is $a_n = a_1 + (n - 1)d$, we have

$$a_{11} = -10 + (11 - 1)(-3) = -10 - 30 = -40.$$

Therefore, the 11^{th} term of the arithmetic sequence we are given is $a_{11} = -40$.

4 pts

9. Write an explicit formula for the geometric sequence

$$\left\{2, 1, \frac{1}{2}, \frac{1}{4}, \dots\right\}$$

Since this is a geometric sequence, we begin by computing the common ratio:

$$r = \frac{1}{2} = \frac{1/2}{1/4} = \frac{1}{2}.$$

Since the explicit formula for a geometric sequence is of the form $a_n = a_1 r^{n-1}$, an explicit formula for the geometric sequence were are given is

$$a_n = 2 \left(\frac{1}{2}\right)^{n-1}.$$

10 pts

10. Find the value of the sum

$$1 + 2 + 3 + \dots + 999 + 1000.$$

Since $\{1, 2, 3, \dots, 999, 1000\}$ is an arithmetic sequence, this is a finite arithmetic series of the form $\sum_{n=1}^{1000} a_n$ with $a_n = n$. So, we use the formula for the sum of the first n terms of an arithmetic series, i.e.

$$S_n = \frac{n(a_1 + a_n)}{2}.$$

Since $n = 1000$, $a_1 = 1$, and $a_{1000} = 1000$ here, we have

$$S_n = \frac{1000(1 + 1000)}{2} = 500(1001) = 500,500.$$

So, we reach our final answer:

$$1 + 2 + 3 + \dots + 999 + 1000 = 500,500.$$

10 pts

11. Find the value of the sum

$$-1 - \frac{1}{4} - \frac{1}{16} - \frac{1}{64} - \dots$$

Since $\{-1, -\frac{1}{4}, -\frac{1}{16}, \dots\}$ is a geometric sequence, this is a geometric series of the form $\sum_{n=1}^{\infty} a_1 r^{n-1}$ with $r = 1/4$. So, since $r < 1$, this series converges to a finite value and we may use the formula for the sum of a geometric series, i.e.

$$\sum_{n=1}^{\infty} a_1 r^{n-1} = \frac{a_1}{1-r}.$$

Since $a_1 = -1$ and $r = 1/4$ here, we have

$$\sum_{n=1}^{\infty} -1 \left(\frac{1}{4}\right)^{n-1} = \frac{-1}{1-(1/4)} = \frac{-1}{3/4} = -\frac{4}{3}.$$

So, we reach our final answer:

$$-1 - \frac{1}{4} - \frac{1}{16} - \frac{1}{64} - \dots = -\frac{4}{3}.$$

10 pts

12. Find the sum of the finite geometric series

$$\sum_{k=1}^{12} 3 \left(\frac{1}{4}\right)^{k-1}$$

Recall that the sum of a finite geometric series is given by

$$\sum_{k=1}^n a_1 (r)^{k-1} = \frac{a_1(1-r^n)}{1-r}.$$

So, the sum of the finite geometric series we are given is

$$\sum_{k=1}^{12} 3 \left(\frac{1}{4}\right)^{k-1} = \frac{3(1-1/4^{12})}{1-1/4} \approx 4$$

10 pts

13. Find the sum of the infinite geometric series

$$\sum_{k=1}^{\infty} -\frac{9}{4} \left(-\frac{4}{9}\right)^{k-1}$$

Recall that the sum of an infinite geometric series with common ratio r such that $-1 < r < 1$ is given by

$$\sum_{k=1}^{\infty} a_1(r)^{k-1} = \frac{a_1}{1-r}.$$

So, since $-1 < -4/9 < 1$, the sum of the infinite geometric series we are given is

$$\sum_{k=1}^{\infty} -\frac{9}{4} \left(-\frac{4}{9}\right)^{k-1} = \frac{-9/4}{1 - (-4/9)} = -\frac{81}{52}$$

- 5 (bonus) 14. Compute the derivative of $f(x) = x^2 - 4$ using using *only* the limit definition of the derivative,

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

Using the definition of the derivative, we calculate:

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{[(x+h)^2 - 4] - [x^2 - 4]}{h} \\ &= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - 4 - x^2 + 4}{h} \\ &= \lim_{h \rightarrow 0} \frac{2xh + h^2}{h} \\ &= \lim_{h \rightarrow 0} 2x + h \\ &= 2x. \end{aligned}$$